

# ELEMENTARY STRATA OF THE CONTINUUM UNDER $\neg$ CH: A LEAN FORMALIZATION STUDY

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**ABSTRACT.** We formalize, in Lean 4 and Mathlib, the elementary-substructure picture of the real continuum under  $\neg$  CH. Downward Löwenheim–Skolem yields elementary substrata of  $\mathbb{R}$  at every intermediate cardinality. Nested elementary substructures of a common model form an elementary pair, so these strata assemble into a strictly increasing  $\mathbb{N}$ -chain and into a monotone covering family of length  $\text{cof}(2^{\aleph_0})$  whose inclusions are elementary. Named real-closed subfields of every infinite cardinality up to the continuum are constructed separately; elementarity of those algebraic inclusions remains conditional on a Mathlib gap (model completeness of RCF). The project declares no axioms: hypotheses appear in type signatures, and kernel audits are machine-checked in CI.

## 1. INTRODUCTION

The Continuum Hypothesis (CH) asserts that there is no cardinal strictly between the cardinality of the natural numbers and the cardinality of the continuum. Consequently, any mathematical program that treats intermediate cardinalities inside  $\mathbb{R}$  must either work relative to  $\neg$  CH, change the meaning of size, or explicitly track which assumptions are external to ZFC.

The Lean development studied here chooses the first route. It introduces a named proposition, `ICAH.NotCH : Prop`, defined as `continuum  $\neq$  aleph 1`, and threads it through the development as an explicit hypothesis. Around this hypothesis, the project builds a formal architecture for elementary substrata of the real continuum, size-aware field structures, named real-closed subfields, elementary chains, and cofinal covering families.

The project name “ICAH” is retained as a repository identifier. The mathematics is standard: downward Löwenheim–Skolem plus Tarski for existence, Tarski–Vaught for chains and directed unions. The contribution is the formalization and the assumption cartography, not a new set-theoretic hypothesis.

### 1.1. Main theorem.

The development is organized as two pillars that realize the same informal picture at different levels of concreteness.

**Theorem. (Theorem 1 — Pillar A, semantic form; Lean: `icahElementary`.)** Assume  $\neg$  CH (i.e. `NotCH`). Then, writing  $\mathcal{L}_{\text{Ring}}$  for the first-order ring language and “stratum” for an elementary substructure  $S \prec \mathbb{R}$  in  $\mathcal{L}_{\text{Ring}}$ :

1. **(band)** there exists a cardinal  $\kappa$  with  $\aleph_0 < \kappa < 2^{\aleph_0}$ ;
2. **(strata at every level)** for every cardinal  $\kappa$  with  $\aleph_0 < \kappa < 2^{\aleph_0}$  there is an elementary substructure  $S \prec \mathbb{R}$  with  $\#S = \kappa$  (this clause is ZFC-pure: downward Löwenheim–Skolem);

3. (**strict chains**) there is a **strictly increasing**  $\mathbb{N}$ -indexed elementary chain of intermediate-size strata whose direct limit is elementarily equivalent to  $\mathbb{R}$ ;
4. (**closure; ZFC-pure**) for every  $\mathbb{N}$ -indexed elementary chain whose levels have size  $< 2^{\aleph_0}$ , the direct limit has size  $< 2^{\aleph_0}$  — countable chains never escape the hierarchy;
5. (**cofinality**) every real number lies in some intermediate-size elementary substructure of  $\mathbb{R}$ .

Separately, and still from  $\neg \text{CH}$  alone, there is a monotone `c.ord`-indexed family of intermediate-size elementary substrata covering  $\mathbb{R}$ , compressible to length  $\text{cof}(2^{\aleph_0})$  (`exists_cofinal_elem_family`, `exists_cofinal_elem_family_cof_length`). Every inclusion is elementary by the nesting lemma.

The Lean proof depends only on the kernel axioms `propext`, `Classical.choice`, `Quot.sound`; this is machine-checked by `#guard_msgs` in the build.

**Theorem. (Theorem 2 — Pillar B, algebraic realization; Lean: `icahTheorem`.)** Assume  $\neg \text{CH}$  and additionally `RCFModelComplete` (compatibility name for the  $\mathbb{R}$ -specialized consequence that the inclusion of every real-closed subfield of  $\mathbb{R}$  into  $\mathbb{R}$  is elementary in the ring language; it follows from model completeness of RCF, a Mathlib gap). Then the clauses of `ICAHStatement` hold:

1. (**M1**) for every ordinal  $n$  there is a stratum  $R$  with index  $n$  and  $\aleph_0 < \#R < 2^{\aleph_0}$ ;
2. (**M3**) every stratum carries a size-aware ordered-field structure with matching carrier and cardinal;
3. (**M5**) there is a **strictly increasing** chain of strata whose direct limit is elementarily equivalent to  $\mathbb{R}$  (the witness is the Pillar A chain; this clause does not use the RCF hypothesis);
4. (**M6**) there is a nonconstant `c.ord`-indexed monotone family of intermediate-size real-closed subfields of  $\mathbb{R}$  whose union is all of  $\mathbb{R}$ ;
5. (**elementary RC family**) under the RCF hypothesis, that same family has elementary inclusions (`exists_cofinal_rc_family_elementary`).

Pillar A is unconditional on model completeness: its strata are elementary by construction, model  $\text{Th}(\mathbb{R})$  (hence the first-order theory of real-closed fields), and assemble into the nonconstant elementary hierarchy the informal picture asks for. Pillar B is the **conditional** pillar in both senses: it supplies named, algebraically explicit real-closed subfields (relative algebraic closures, with native `IsRealClosed` instances), and it purchases elementarity of those algebraic inclusions through the remaining Mathlib gap. Once a ring-language RCF axiomatization is transferred to a native `IsRealClosed` instance, Pillar A already supplies concrete-enough real-closed elementary strata without `RCFModelComplete`.

**Remark.** Three constructions must not be conflated. First, `tarskiVaughtDirectLimit` and `ofLevelElem` are general theorems about arbi-

trary  $\aleph_1$ -indexed elementary chains. Second, the existential chain clauses of `icahElementary` and `icahTheorem` are discharged by the **strictly increasing** chain `strictElemChain`, built by iterating downward Löwenheim–Skolem from an  $\aleph_1$ -sized substratum and adjoining a fresh real at each step; successor maps are the nesting inclusions. Third, `exists_cofinal_elem_family` is a genuinely nonconstant monotone covering family of elementary substrata of length at most  $c$ , compressible to  $\text{cof}(c)$ . The older constant-chain objects remain in the library as test cases; they are not the witnesses of the top-level theorems.

**Contribution.** The main contribution is not new mathematics — to a model theorist, the existence of elementary (hence first-order real-closed) substrata of every intermediate cardinality is a corollary of downward Löwenheim–Skolem, and the chain and union results are textbook Tarski–Vaught. The contribution is the formalization architecture and the assumption cartography: a Lean-readable decomposition into proved lemmas, explicitly threaded hypotheses, and Mathlib-facing gaps, with the dependency set of every flagship theorem machine-audited in the build. The development declares **zero** axioms; the semantic theorem (Pillar A) holds under  $\neg \text{CH}$  alone.

The project should be positioned for a mathematical audience as a formalization paper with three layers:

1. a set-theoretic layer, where the hypothesis  $\neg \text{CH}$  produces intermediate cardinal witnesses;
2. a model-theoretic/algebraic layer, where elementary substrata of every intermediate cardinality are produced by downward Löwenheim–Skolem (Pillar A), general elementary-chain theorems are proved, a cofinal elementary family of optimal length is constructed, and named real-closed subfields of every infinite cardinality up to the continuum are constructed (Pillar B);
3. a proof-engineering layer, where the remaining Mathlib-facing dependency of Pillar B is exposed as a named hypothesis, machine-audited in the build, and can be attacked as an independent Mathlib contribution.

The paper is organized as follows. Section 2 recalls the mathematical background. Section 3 describes the Lean architecture. Section 4 isolates the results that are already proved. Section 5 gives the hypothesis inventory, the per-theorem audit, and the Mathlib roadmap. Section 6 discusses related work, including the formalizations of the independence of CH and the certified quantifier-elimination procedures for real-closed fields in other proof assistants. Section 7 briefly notes analogues in valued and p-adic field theory. Section 8 concludes with reproducibility data and an engagement plan.

## 1.2. Guiding problem.

**Definition.** An *intermediate-cardinality stratum* is, informally, a subset  $S \subseteq \mathbb{R}$  equipped with a cardinal  $\kappa$  such that

$$\aleph_0 < \kappa < 2^{\aleph_0}$$

and a proof that the subtype determined by  $S$  has cardinality  $\kappa$ .

**Definition.** A *size-aware field* is a bundled field-like object carrying its underlying type, a designated cardinal, and the typeclass instances needed to treat the carrier as an ordered ring or field in Lean.

The central formal question is:

**Question.** Which components of an intermediate-cardinality stratification can be formalized under explicit assumptions, what general elementary-chain theorems are available, and how long must a family of intermediate-size elementary substrata or real-closed subfields be to exhaust  $\mathbb{R}$ ?

The current Lean code gives complete proofs of the stated `ICAHElementary` clauses under  $\neg \text{CH}$  and of the stated `ICAHStatement` clauses under its two explicit hypotheses. The chain witnesses are strictly increasing; the cofinal elementary family of length  $\text{cof}(2^{\aleph_0})$  is proved. For arbitrary covering families of subsets of  $\mathbb{R}$  whose members have size below the continuum, the lower bound is  $\text{cof}(2^{\aleph_0})$ ; under  $\neg \text{CH}$ , this bound is attained both by elementary substrata and by named real-closed subfields.

## 2. BACKGROUND

### 2.1. The continuum and the role of $\neg \text{CH}$ .

Let  $c = 2^{\aleph_0}$  denote the cardinality of the continuum. The Continuum Hypothesis states that there is no cardinal  $\kappa$  satisfying  $\aleph_0 < \kappa < c$ . Since Gödel and Cohen, the standard mathematical background is that CH is independent of ZFC, assuming ZFC is consistent [1, 2].

The Lean project therefore does not attempt to prove  $\neg \text{CH}$  in ZFC. Instead, it defines  $\neg \text{CH}$  as a named proposition (`ICAH.NotCH`) and threads it as an explicit hypothesis through every theorem that needs it. The environment contains no project axioms; the per-theorem `#print axioms` audit reports only the Lean kernel axioms.

**Proposition.** Under the hypothesis `ICAH.NotCH`, the cardinal  $\aleph_1$  witnesses the existence of an intermediate cardinal:

$$\exists \kappa, \aleph_0 < \kappa \wedge \kappa < 2^{\aleph_0}.$$

The Lean proof uses the standard inequality  $\aleph_0 < \aleph_1$ , the theorem  $\aleph_1 \leq c$ , and converts the inequality plus non-equality  $c \neq \aleph_1$  into the strict inequality  $\aleph_1 < c$ .

### 2.2. Real closed fields and definability.

Real closed fields are the algebraic and model-theoretic setting closest to the ordered real field. They admit quantifier elimination [3] and provide the natural language in which order, addition, multiplication, and definability can be studied together [4, 5].

ICAH uses this background in a restrained way. The code does not merely assume every bare subset of  $\mathbb{R}$  is a field. Instead, it introduces `SubfieldStratum`, which requires the carrier set to be the underlying set of a `Subfield RR`. This is

the right formal move: closure under field operations is obtained from the subfield structure rather than reconstructed ad hoc.

### 2.3. Elementary substructures and downward Löwenheim–Skolem.

An elementary substructure  $S \prec M$  satisfies the same first-order formulas as  $M$  at parameters from  $S$ , by definition. The downward Löwenheim–Skolem theorem produces such substructures at every prescribed infinite cardinality between the language size and  $\#M$  [6]. Mathlib formalizes this as `exists_elementarySubstructure_card_eq` (via Skolem functions), and Pillar A of the present development is essentially a specialization of that theorem to  $\mathbb{R}$  in the ring language: elementarity of the strata is free by construction.

### 2.4. Elementary chains and direct limits.

An elementary chain is a sequence of structures connected by elementary embeddings. The Tarski–Vaught elementary chain theorem asserts that the union (or directed limit) of such a chain is an elementary extension of each level [6, 7].

The Lean project packages this through `ElemChain`, where each level carries an  $\mathcal{L}_{\text{Ring}}$  structure and each successor map is an elementary embedding. The direct limit is then implemented using Mathlib’s `Language.DirectLimit` API. The project proves the chain theorem **relativized to the ambient model  $\mathbb{R}$**  (`tarskiVaughtDirectLimit`); an earlier draft misattributed this result to Łoś, whose theorem concerns ultraproducts — the misnomer has been corrected throughout. A companion lemma (`directed_iSup_isElementary`) records the Tarski–Vaught test for directed unions of elementary substrata of  $\mathbb{R}$ : an existential witnessed in  $\mathbb{R}$  with parameters from the union has its parameters in some member, which is already  $\prec \mathbb{R}$ .

## 3. FORMAL ARCHITECTURE

This section records the paper-level interpretation of the Lean objects. The goal is to make the repository legible to mathematicians without forcing them to read Lean code first.

The development is organized as two pillars sharing the cardinal scaffolding:

1. **Pillar A** (`ICAH/ElementaryStrata.lean`): strata are elementary substructures of  $\mathbb{R}$  in the ring language, produced by downward Löwenheim–Skolem. Elementarity is definitional; the main theorem `icahElementary` needs only `NotCH`. Nested elementary substructures of  $\mathbb{R}$  form an elementary pair, so the strata assemble into a strictly increasing  $\mathbb{N}$ -chain and a cofinal elementary family.
2. **Pillar B** (`ICAH/FieldOnStratum.lean` and downstream): strata are concrete real-closed subfields (relative algebraic closures of generated subfields), with native `IsRealClosed` instances. Elementarity of those **named algebraic** inclusions is purchased through `RCFModelComplete`. Pillar B is the conditional pillar.

### 3.1. Core objects.

**Definition.** `Stratum` is the basic continuum-layer object. It consists of an ordinal index, a set  $S \subseteq \mathbb{R}$ , a cardinal  $\kappa$ , a witness that the subtype determined by  $S$  has cardinality  $\kappa$ , and bounds  $\aleph_0 < \kappa < c$ .

**Definition.** `SizeAwareField` is a bundled structure containing a carrier type, a designated cardinal, and the instances `[Field K]`, `[LinearOrder K]`, and `[IsStrictOrderedRing K]`. This avoids relying on deprecated or overly strong ordered-field classes while still exposing enough structure for model-theoretic use.

**Definition.** `SubfieldStratum` refines `Stratum` by adding a subfield `subfield : Subfield RR` and a proof that the stratum set is exactly the underlying set of that subfield.

The role of `SubfieldStratum` is central. A raw set of real numbers is not automatically closed under addition, multiplication, negation, and inverses. A subfield is. Thus, the field part of the theory is moved from a fragile closure proof obligation into a stable algebraic structure.

**Definition.** `RCSubfieldStratum` refines `SubfieldStratum` once more by requiring the subfield to be **real closed**. This is the correct hypothesis for the algebraic pillar: a general subfield of  $\mathbb{R}$  (such as  $\mathbb{Q}$ ) is not elementarily embedded in  $\mathbb{R}$ , while the inclusion of a real-closed subfield into  $\mathbb{R}$  is elementary by the specialized consequence of model completeness used here.

**Definition.** In Pillar A, a stratum is an `LRing.ElementarySubstructure ℝ` — Mathlib’s bundled elementary substructure. The theorem `exists_elementary_substratum` produces one of every infinite cardinality  $\kappa \leq 2^{\aleph_0}$  (optionally containing a prescribed set of size  $\leq \kappa$ , via `exists_elementary_substratum_extending`), by instantiating Mathlib’s downward Löwenheim–Skolem theorem `exists_elementarySubstructure_card_eq` at  $M = \mathbb{R}$ . The bridge lemma `elemSubstratumSubfield` shows every such stratum is the carrier of a subfield of  $\mathbb{R}$ : closure under ring operations is the substructure property, and closure under inverses is one formula transfer ( $\exists y, x \cdot y = 1$ ) along elementarity. First-order real-closedness is immediate: an elementary substratum models  $\text{Th}(\mathbb{R})$  (`elemSubstratum_models_thReal`), and  $\text{Th}(\mathbb{R})$  contains the ring-language axiomatization of RCF. Native `IsRealClosed` instance transfer along this observation is not formalized.

**Construction.** `subfieldToSAF` converts a subfield of  $\mathbb{R}$ , together with a cardinality witness, into a `SizeAwareField`. The key design choice is to use subtype inheritance for the linear order and Mathlib’s subfield ordered-ring instance for the strict ordered ring structure.

**Construction.** `relAlgebraic K₀` is the relative algebraic closure of a subfield  $K_0 \subseteq \mathbb{R}$  inside  $\mathbb{R}$ . It is proved real closed via a root-closure criterion, and its cardinality is bounded by the maximum of the cardinality of  $K_0$  and  $\aleph_0$ . This is the engine behind `exists_rc_subfield`, which produces a real-closed subfield of any prescribed infinite cardinality up to the continuum.

### 3.2. Language.

The language is Mathlib's `Language.ring`, abbreviated `LRing` (historical compatibility name `LOR`): function symbols  $+$ ,  $\cdot$ ,  $-$ ,  $0$ ,  $1$  and `no` relation symbols. It is not a language of ordered rings. This costs nothing for the intended models, because in a real-closed field the order is definable from the ring structure ( $x \leq y$  iff  $y - x$  is a square). The graphs of addition and multiplication are therefore atomic formulas; the corresponding definability lemmas are API stress tests of Mathlib's realization machinery, not mathematical content, and are not given a separate subsection.

### 3.3. Elementary chains.

**Definition.** `ElemChain` packages a sequence `obj : ℕ -> Type*` of first-order structures and elementary embeddings `obj n ↪ₑ obj (n+1)`.

From these successor maps, the project defines two related systems:

1. `embLE`, which composes successor elementary embeddings and preserves elementarity;
2. `sysEmb`, which uses Mathlib's directed-system API to build the underlying embedding system.

The lemma `embLE_eq_sysEmb` proves that these two constructions agree as functions. This is a small but important bridge: the proof-relevant elementary embedding API and the directed-colimit API are not automatically the same object.

**Lemma.** `elementaryInclusion`: if  $S \subseteq T$  are both elementary substructures of a common model  $M$ , then the inclusion  $S \rightarrow T$  is elementary. Satisfaction in  $S$  and in  $T$  both reduce to satisfaction in  $M$ .

**Construction.** `strictElemChain`: under `NotCH`, start from an  $\aleph_1$ -sized elementary substratum of  $\mathbb{R}$  (DLS); at each stage pick a real outside the current carrier (possible because the carrier has size  $< c$ ) and apply `exists_elementary_substratum_extending` to the previous carrier plus that real. Successor maps are the nesting inclusions. The result is a strictly increasing  $\mathbb{N}$ -indexed elementary chain of intermediate-size strata; `ofLevelElem` and `directLimit_card_lt_continuum` handle the limit.

### 3.4. Direct limit.

**Definition.** For an elementary chain  $C$ , the direct limit `DirectLim C` is defined as `Language.DirectLimit C.obj (...)`, using the underlying directed system of embeddings.

The direct limit is the formal version of the intended limit field  $F_\omega$ . The project proves the cardinality identity  $\#F_\omega = \sup_n \#C_n$  (`directLimit_card_eq_iSup`), the closure theorem that countable chains of intermediate strata stay intermediate (`directLimit_card_lt_continuum`, via König), and two elementarity theorems: the relativized version over  $\mathbb{R}$  (`tarskiVaughtDirectLimit`, via the Tarski–Vaught test) and the ambient-free version (`ofLevelElem`: the canonical maps into the direct limit are elementary, by induction on bounded formulas). Both elementarity statements were previously exposed as gaps and are now theorems.

### 3.5. Cofinal families at the continuum, and optimality.

The  $\mathbb{N}$ -indexed chain cannot, by itself, exhaust  $\mathbb{R}$ : König’s theorem gives  $\text{cof}(c) > \aleph_0$ , so a countable increasing union of sets of size  $< c$  has size  $< c$ . The module `ICAH/CofinalFamily.lean` therefore constructs two  $\iota.\text{ord}$ -indexed monotone covering families, both compressible to length  $\text{cof}(c)$ :

1. `exists_cofinal_elem_family`: elementary substrata, seeded at each ordinal by the previous stage plus the next real under an enumeration of  $\mathbb{R}$ . Inclusions are elementary by nesting; limit stages are elementary by the directed-union lemma. This family needs only `NotCH`.
2. `exists_cofinal_rc_family`: named real-closed subfields (relative algebraic closures). Under `RCFModelComplete`, its inclusions are elementary (`exists_cofinal_rc_family_elementary`).

The module also proves a qualified exact threshold: any family of subsets of  $\mathbb{R}$ , each of size  $< c$ , covering  $\mathbb{R}$  has at least  $\text{cof}(c)$  members (`cofinal_family_length_lower_bound`). See Section 4.

## 4. PROVED RESULTS

This section separates fully proved Lean results from hypotheses. Section 8 contains the complete declaration-to-paper mapping table; Section 5 records, for each result, which of the two named hypotheses (if any) it consumes.

### 4.1. Intermediate cardinal witness.

**Theorem.** `exists_intermediate_cardinal`: assuming the hypothesis `NotCH`, there exists a cardinal  $\kappa$  such that  $\aleph_0 < \kappa < c$ .

**Remark.** The witness is  $\aleph_1$ . The proof uses `aleph0_lt_aleph_one`, `aleph_one_le_continuum`, and turns non-equality into strict inequality (`aleph_one_lt_continuum_of_notCH`).

### 4.2. Elementary substrata via downward Löwenheim–Skolem (Pillar A).

**Theorem.** `exists_elementary_substratum_extending` (ZFC-pure): for every cardinal  $\aleph_0 \leq \kappa \leq c$  and every set  $s \subseteq \mathbb{R}$  with  $\#s \leq \kappa$ , there is an elementary substructure  $S \prec \mathbb{R}$  in the ring language with  $s \subseteq S$  and  $\#S = \kappa$ .

**Remark.** This is Mathlib’s downward Löwenheim–Skolem theorem `exists_elementarySubstructure_card_eq` instantiated at  $M = \mathbb{R}$ ; the language has five symbols (`card_ring`), so the language-size condition is absorbed by

$\aleph_0 \leq \kappa$ . Elementarity of the strata is free — no Tarski–Seidenberg. This single theorem powers every clause of `icahElementary` (Theorem 1).

**Theorem.** `elemSubstratumSubfield`: the carrier of any elementary substructure of  $\mathbb{R}$  in the ring language is a subfield of  $\mathbb{R}$ .

**Remark.** Closure under  $+, \cdot, -, 0, 1$  is the substructure property; closure under inverses is a transfer of the formula  $\exists y, x \cdot y = 1$  along elementarity. This is the satisfaction-to-structure bridge that gives Pillar A strata an algebraic identity.

**Theorem.** `elemSubstratum_models_thReal` (ZFC-pure): an elementary substratum of  $\mathbb{R}$  satisfies every sentence true in  $\mathbb{R}$ . In particular it models  $\text{Th}(\mathbb{R})$ , which contains the first-order axiomatization of real-closed fields in  $\mathcal{L}_{\text{Ring}}$ .

**Remark.** RCF is a first-order theory in the ring language (order defined via squares; the axiom schema is the ring axioms, every nonnegative element is a square, and odd-degree polynomials have roots). An elementary substructure of  $\mathbb{R}$  therefore satisfies RCF and is real closed in the first-order sense. Native Lean `IsRealClosed` instance transfer along this observation is not formalized; that is a packaging gap, not a first-order gap. Pillar B remains the place where **named** real-closed subfields carry `IsRealClosed` instances.

#### 4.3. Nesting, directed unions, and a strictly increasing chain.

**Lemma.** `elementaryInclusion` (ZFC-pure): if  $S \subseteq T$  are both elementary substructures of a common model  $M$ , then  $S \prec T$ . Satisfaction in  $S$  and in  $T$  both reduce to satisfaction in  $M$ .

**Theorem.** `directed_iSup_isElementary` (ZFC-pure): the directed supremum of a nonempty family of elementary substructures of  $\mathbb{R}$  is again elementary in  $\mathbb{R}$ . Specialization: the union of a nonempty monotone chain is elementary (`iUnionChainElem`).

**Remark.** This is the Tarski–Vaught test verbatim: an existential witnessed in  $\mathbb{R}$  with parameters from the union has its (finitely many) parameters in some common member, which is already  $\prec \mathbb{R}$ .

**Construction.** `strictElemChain` (under `NotCH`): a strictly increasing  $\aleph$ -indexed elementary chain of  $\aleph_1$ -sized substrata of  $\mathbb{R}$ . Start from DLS at  $\aleph_1$ ; since  $\aleph_1 < c$ , pick  $x \notin S_n$ ; apply `exists_elementary_substratum_extending` to  $S_n \cup \{x\}$ . Successor maps are nesting inclusions. The direct limit is elementarily equivalent to  $\mathbb{R}$  (`strictElemChain_directLim_equiv`).

This construction is the witness of the chain clauses in both top-level theorems.

#### 4.4. Concrete synthetic stratum.

**Construction.** `syntheticStratum`: under `NotCH`, materializes an intermediate-cardinality subset of  $\mathbb{R}$  by extracting a concrete set  $S \subseteq \mathbb{R}$  with cardinality  $\aleph_1$  from a cardinal existence theorem.

This is the first bridge from a cardinal statement to an inhabited Lean structure. It turns the abstract existence of an intermediate cardinal into a concrete `Stratum` object.

#### 4.5. Subfield strata and size-aware fields.

**Theorem.** `fieldOnSubfieldStratum`: every `SubfieldStratum` yields a `SizeAwareField` whose carrier and cardinal match the underlying stratum.

The technical point is that the carrier equality is treated as a propositional equality of subtypes, not merely a bijection. This is precisely what the `SizeAwareField.carrier` field requires.

**Lemma.** `fieldOnStratum`: every `Stratum` (not only subfield strata) admits a `SizeAwareField` with matching carrier and cardinal. Formerly an axiom.

**Remark.** This is deliberately stated as a lemma, not a theorem: it is a **packaging** result. The proof transports the field structure of a real-closed subfield of the right cardinality (from `exists_rc_subfield`, below) across a cardinality `Equiv`, using `Equiv.field` and `LinearOrder.lift'` — so the resulting operations on the carrier are an arbitrary transport and bear no relation to the ambient operations of  $\mathbb{R}$ . The mathematically meaningful field structures live on `SubfieldStratum` and `RCSubfieldStratum`, whose operations **are** those of  $\mathbb{R}$ .

#### 4.6. Real-closedness of $\mathbb{R}$ and of relative algebraic closures.

**Theorem.** `Real.isRealClosed`:  $\mathbb{R}$  is a real-closed field. Formerly an axiom.

**Remark.** The proof uses `IsRealClosed.of_linearOrderedField`: squares are handled by `Real.sqrt`, and odd-degree polynomials acquire roots by the intermediate value theorem combined with the leading-coefficient asymptotics `Polynomial.tendsto_atTop/atBot`.

**Theorem.** `isRealClosed_of_forall_root` and `relAlgebraic_isRealClosed`: a subfield of  $\mathbb{R}$  closed under taking real roots of its polynomials is real closed; in particular the relative algebraic closure `relAlgebraic K₀` of any subfield  $K_0 \subseteq \mathbb{R}$  is real closed.

**Theorem.** `exists_rc_subfield` (ZFC-pure): for every cardinal  $\kappa$  with  $\aleph_0 \leq \kappa \leq c$ , there exists a real-closed subfield of  $\mathbb{R}$  of cardinality exactly  $\kappa$ .

**Remark.** This proves (and strengthens) the former `subfieldStratumExists` axiom. The subfield is the relative algebraic closure of the sub-

field generated by a set of size  $\kappa$ ; the cardinality is pinned by `Algebra.IsAlgebraic.cardinalMk_le_max` for the upper bound and injectivity of the generators for the lower bound. It is the concrete (Pillar B) counterpart of `exists_elementary_substratum`: an explicit algebraic description, at the price that elementarity over  $\mathbb{R}$  is no longer free.

#### 4.7. Real algebraic numbers as a base field.

**Theorem.** `algReal_card_le_aleph0`: the real algebraic numbers, represented as `algebraicClosure QQ RR` inside  $\mathbb{R}$ , have cardinality at most  $\aleph_0$ .

Together with the lower bound for infinite types, this supports the concrete base object `algRealSAF`, a countable size-aware field outside the intermediate-cardinality hierarchy.

#### 4.8. Elementarity-preserving composition.

**Theorem.** `embLE`: for any  $m \leq n$ , the chain induces an elementary embedding from level  $m$  to level  $n$ .

**Theorem.** `embLE_eq_sysEmb`: the elementarity-preserving embedding built by recursive composition agrees as a function with the directed-system embedding used by `Language.DirectLimit`.

#### 4.9. Tarski–Vaught theorem for direct limits.

**Theorem.** `ElemChain.tarskiVaughtDirectLimit`: if every level of an elementary chain embeds elementarily and compatibly into  $\mathbb{R}$ , then the direct limit is elementarily equivalent to  $\mathbb{R}$ . Formerly an axiom.

**Remark.** The underlying map is built by `Language.DirectLimit.lift`; elementarity is established by the Tarski–Vaught test: a witness of an existential in  $\mathbb{R}$  over parameters from the limit is pulled back to a finite level via the compatibility of the embeddings. **Naming note:** an earlier draft called this result “Łoś-style”; that was a misattribution (Łoś’s theorem concerns ultraproducts; the relevant classical result is the Tarski–Vaught elementary chain theorem, here relativized to the ambient model  $\mathbb{R}$ ).

**Theorem.** `ElemChain.ofLevelElem` and `realize_ofLevel_iff` (ZFC-pure; **ambient-free** Tarski–Vaught chain theorem): for any elementary chain, the canonical embedding of each level into the direct limit is itself **elementary** — no ambient model required. Consequently the direct limit is elementarily equivalent to every level (`directLim_elementarilyEquivalent`).

**Remark.** Unlike the relativized version, this cannot be obtained from Mathlib’s Tarski–Vaught test (pulling realization back through `DirectLimit.of` for arbitrary formulas is precisely what is being proved). The proof is a direct induction

on `BoundedFormula`; in the `all` case, a direct-limit witness is pulled down to a common level via directedness and `DirectLimit.of_f`, while the universally quantified hypothesis is pushed up through the elementary transition embeddings. Mathlib’s `DirectLimit` file currently has no elementarity results at all; this theorem is the form Mathlib should receive.

#### 4.10. Cardinality of the direct limit: identity, closure, and sharpness.

An earlier draft stated `directLimit_card`: if every level has cardinality below the continuum **and** the supremum of the level cardinalities is the continuum, **then** the direct limit has cardinality continuum. Those hypotheses are jointly unsatisfiable in ZFC — König’s theorem gives  $\text{cof}(c) > \aleph_0$ , so a countable supremum of cardinals  $< c$  stays  $< c$  — making the theorem true and empty. It has been deleted and replaced by three non-vacuous results.

**Theorem.** `directLimit_card_eq_iSup` (ZFC-pure): for an  $\mathbb{N}$ -indexed chain with infinite supremum of level cardinalities,  $\#(\text{DirectLim } C) = \sup_n \#C_n$ .

The proof has two halves. The upper bound exhibits the direct limit as a quotient of the sigma type  $\sum_{n:\mathbb{N}} C_n$ , bounding its cardinality by  $\aleph_0 \cdot \sup_n \#C_n = \sup_n \#C_n$ . The lower bound uses injectivity of the canonical maps from each level into the direct limit and then takes the supremum over levels.

**Theorem.** `directLimit_card_lt_continuum` and `directLimit_intermediate` (ZFC-pure; closure): if every level of an  $\mathbb{N}$ -indexed chain has cardinality  $< c$ , so does the direct limit; if moreover the first level is uncountable, the limit is again of intermediate size. Countable elementary chains never escape the stratum hierarchy.

**Theorem.** `cofinal_family_length_lower_bound` (ZFC-pure; sharpness, lower bound): any family  $\{S_i\}_{i \in \iota}$  of subsets of  $\mathbb{R}$  with  $\#S_i < c$  for all  $i$  and  $\bigcup_i S_i = \mathbb{R}$  satisfies  $\#\iota \geq \text{cof}(c)$ .

#### 4.11. Cofinal elementary and real-closed families.

**Theorem.** `exists_cofinal_elem_family` (under `NotCH`): there is a monotone,  $\aleph_1$ -ord-indexed family of elementary substructures of  $\mathbb{R}$ , each of intermediate cardinality, covering  $\mathbb{R}$ . Every inclusion is elementary by `elementaryInclusion`. Compressing along a fundamental sequence yields a covering family of length exactly  $\text{cof}(c)$  (`exists_cofinal_elem_family_cof_length`).

**Theorem.** `exists_cofinal_rc_family` and `exists_cofinal_rc_family_cof_length` (under `NotCH`): the same cardinal picture for **named** real-closed subfields (relative algebraic closures). Under `RCFModelComplete`, the inclusions of that family are elementary (`exists_cofinal_rc_family_elementary`, via `nestedRCEmbedding`).

**Remark.** Together with the lower bound, these results say: among covering families whose members are subsets of  $\mathbb{R}$  of size  $< c$ , the minimum possible index cardinal is  $\text{cof}(c)$ ; under **NotCH**, that minimum is attained both by elementary substrata and by named real-closed subfields. The  $\aleph_N$ -indexed chain cannot exhaust  $\mathbb{R}$  (König); the cofinal families can.

**Contribution.** For the paper, `exists_elementary_substratum`, `elementaryInclusion`, `directed_iSup_isElementary`, `strictElemChain`, `exists_cofinal_elem_family`, `ofLevelElem`, `directLimit_card_eq_iSup`, the  $\text{cof}(c)$  sharpness pair, and `exists_rc_subfield` are the strongest fully proved results to foreground. Each is independent of any grand interpretation of a named hypothesis, and all but the **NotCH**-consuming ones are ZFC-pure — which is precisely their selling point as Mathlib upstreaming candidates.

## 5. HYPOTHESIS INVENTORY, PER-THEOREM AUDIT, AND MATHLIB ROADMAP

A central strength of the Lean development is that it does not hide its assumptions. The project declares **zero** axioms: the two named assumptions are ordinary **Props** threaded through the development as explicit hypotheses, so the assumption set of every theorem is visible in its type signature. The kernel-level audit is enforced in-source by `#guard_msgs` in `#print axioms` blocks and checked in CI:

```
'ICAH.icalElementary' depends on axioms:
  [propext, Classical.choice, Quot.sound]
'ICAH.icalTheorem' depends on axioms:
  [propext, Classical.choice, Quot.sound]
```

Any drift in these sets — including a hidden `sorryAx` — fails the build. With hypotheses in signatures rather than axioms in the environment, `#guard_msgs` is demoted from safety mechanism to regression test, which is the right division of labor.

### 5.1. External mathematical hypothesis.

**Assumption.** `ICAH.NotCH` : **Prop**, defined as `continuum ≠ aleph 1`: the negation of the Continuum Hypothesis.

This is not a Mathlib gap. It is the intended set-theoretic regime of the project: the theory is developed relative to  $\neg \text{CH}$ , which is consistent with ZFC by Cohen’s forcing argument [2] — itself formalized in Lean by the Flypitch project [8] (see Section 6).

### 5.2. The remaining Mathlib-facing gap (Pillar B).

**Mathlib gap.** `ICAH.RCFSubfieldRealElementary` : **Prop** (compatibility spelling `ICAH.RCFModelComplete`): the inclusion of a real-closed subfield of  $\mathbb{R}$  into  $\mathbb{R}$  is an elementary embedding in the ring language (in which the order of a real-closed field is definable via squares). This is the  $\mathbb{R}$ -specialized consequence of model completeness of the theory of real-closed fields needed by Pillar B.

This is a true classical theorem whose first-order formalization follows from quantifier elimination [3] or model completeness for RCF inside Mathlib’s `ModelTheory`

framework, neither of which is currently available in the needed form in the pinned Mathlib revision. It is consumed only by Pillar B (`nestedRCEmbedding`, `exists_cofinal_rc_family_elementary`, and the last clause of `icahTheorem`). Pillar A (`icahElementary`, `exists_cofinal_elem_family`) avoids it entirely via downward Löwenheim–Skolem.

The claim that this is “the remaining gap” is local to this development and the pinned Mathlib revision. Ordered-field model theory is an active area; the engagement plan in Section 8 posts a minimized statement on the Lean Zulip **before** any upstream PR, precisely to catch in-progress work.

**Remark. Why this gap is harder than the ACF precedent.** Mathlib already contains the deep-embedded model theory of algebraically closed fields: the theory  $\text{ACF}_p$  over the ring language, its completeness for  $p$  prime or zero (`ACF_isComplete`), and the Lefschetz principle. That suggests a template — but completeness of ACF comes cheap via uncountable categoricity and the Łoś–Vaught test, a route that is **closed** for RCF: the theory of real-closed fields is unstable (it defines a linear order) and has the maximum number of models in every uncountable cardinality. Discharging `RCFSubfieldRealElementary` therefore requires QE-grade work — e.g. Robinson’s model-completeness test with sign-change/root-counting embedding arguments inside the deep embedding — substantially more than porting the ACF files.

Prior art calibrates the effort: the `math-comp/real-closed` library in Coq/Rocq contains a certified quantifier-elimination procedure for RCF with the decision procedure `rcf_sat` and its correctness proof (Cohen–Mahboubi [9]), and HOL Light has McLaughlin–Harrison’s proof-producing decision procedure for RCF [10]. Neither transfers directly to Mathlib’s `FirstOrder` framework, but Cohen–Mahboubi is the closest blueprint.

### 5.3. Per-theorem audit.

Because the hypotheses are explicit, the audit can be given per theorem. The classification below is the contractual content of the development:

1. **ZFC-pure** (kernel axioms only, no project hypotheses): `exists_elementary_substratum`, `exists_elementary_substratum_extending`, `elemSubstratumSubfield`, `elemSubstratum_models_thReal`, `elementaryInclusion`, `directed_iSup_isElementary`, `exists_rc_subfield`, `relAlgebraic_isRealClosed`, `isRealClosed_of_forall_root`, `Real.isRealClosed`, `tarskiVaughtDirectLimit`, `ofLevelElem` (the ambient-free chain theorem), `directLimit_card_eq_iSup`, `directLimit_card_lt_continuum`, `directLimit_intermediate`, `cofinal_family_length_lower_bound`, `fieldOnStratum`, `fieldOnSubfieldStratum`, `algReal_card_le_aleph0`.
2. **Under NotCH**: `exists_intermediate_cardinal`, `syntheticStratum`, `subfieldStratumExists`, `intermediateRCSubfieldStratum`, `strictElemChain`, `exists_cofinal_elem_family`, `exists_cofinal_elem_family_cof_length`, `exists_cofinal_rc_family`,

`exists_cofinal_rc_family_cof_length`, `cofinal_family_limit_size`, and the Pillar A main theorem `icahElementary`.

3. **Under NotCH and RCFModelComplete:** `nestedRCEmbedding`, `exists_cofinal_rc_family_elementary`, and the last clause of `icahTheorem`. The M1/M3/M5/M6 clauses of `icahTheorem` are available from `NotCH` alone; the RCF hypothesis is used only for elementary inclusions of the named real-closed family.

The ZFC-pure results need neither hypothesis — that property is precisely their upstream selling point.

#### 5.4. Resolved former axioms.

The earlier inventory was discharged in two stages. First, five axioms became theorems:

**Theorem.** `ICAH.Real.isRealClosed` — **proved** (`ICAH/RealClosed.lean`):  $\mathbb{R}$  is real closed, via `IsRealClosed.of_linearOrderedField`, `Real.sqrt` for squares, and the intermediate value theorem for odd-degree polynomials.

**Theorem.** `ICAH.fieldOnStratum` — **proved** (`ICAH/FieldOnStratum.lean`): every stratum admits a size-aware field, by `Equiv-transport` of the field structure of a real-closed subfield of matching cardinality (a packaging lemma; see Section 4).

**Theorem.** `ICAH.subfieldStratumExists` — **proved and strengthened** (`exists_rc_subfield`): for every  $\aleph_0 \leq \kappa \leq c$  there is a **real-closed** subfield of  $\mathbb{R}$  of cardinality exactly  $\kappa$ , obtained as the relative algebraic closure of a generated subfield.

**Theorem.** `ICAH.ElemChain.loDirectLimit` — **proved and renamed** (`tarskiVaughtDirectLimit`, `ICAH/ElementaryChain.lean`): the direct limit of an elementary chain compatibly embedded in  $\mathbb{R}$  is elementarily equivalent to  $\mathbb{R}$ , via `Language.DirectLimit.lift` and the Tarski–Vaught test.

Second, the two remaining axioms (`not_CH`, `rcfModelComplete`) were converted from axioms into the explicit hypotheses `NotCH` and `RCFSubfieldRealElementary` (with compatibility spelling `RCFModelComplete`), emptying the axiom inventory entirely.

**Remark.** Two further axioms of the earlier draft, `subfieldIsRealClosed` and `subfieldStratumElemEmb`, were **deleted as false in the stated generality**:  $\mathbb{Q}$  is a subfield of  $\mathbb{R}$  that is neither real closed nor elementarily embedded (the sentence  $\exists x, x^2 = 2$  distinguishes  $\mathbb{Q}$  from  $\mathbb{R}$ ). They are replaced by the sound refinement `RCSubfieldStratum` (a stratum whose carrier is a real-closed subfield), for which inclusion elementarity follows from `RCFModelComplete`. This self-correction record — two former assumptions identified as false, with counterexample, and repaired by an honest weakening — is, we believe, the most scientifically credible feature of the methodology, and the reason the audit infrastructure exists.

#### 5.5. The honest limit construction.

The original  $\mathbb{N}$ -indexed chain formulation of the limit-size milestone is vacuous: König’s theorem gives  $\text{cof}(c) > \aleph_0$ , so no countable chain of intermediate-size strata can union to  $\mathbb{R}$ . The same disease affected the earlier cardinality theorem `directLimit_card`, whose hypotheses were jointly unsatisfiable; it has been replaced by the identity/closure/sharpness triple of Section 4. The corrected limit-size statements index families by `ι.ord` (or by  $\text{cof}(c)$  after compression along a fundamental sequence):

**Theorem.** `exists_cofinal_elem_family`: a monotone family of elementary substrata of  $\mathbb{R}$ , indexed by `ι.ord`, each of intermediate cardinality, covering  $\mathbb{R}$ , with elementary inclusions. The same length optimum  $\text{cof}(c)$  is attained (`exists_cofinal_elem_family_cof_length`). The named real-closed counterpart is `exists_cofinal_rc_family`; under `RCFModelComplete` its inclusions are elementary.

### 5.6. Recommended order of attack.

1. Post the minimized in-framework statement of `RCFSubfieldRealElementary` on the Lean Zulip model-theory stream (resolving possible overlap with in-progress ordered-field/real-closure work) before any PR.
2. PR 1 (small, ZFC-pure, attractive): the DLS-based elementary-substrata existence for  $\mathbb{R}$  plus the direct-limit cardinality identity `directLimit_card_eq_iSup`.
3. PR 2: the ambient-free Tarski–Vaught chain theorem for `Language.DirectLimit` (proved here as `ofLevelElem / realize_ofLevel_iff` for  $\mathbb{N}$ -indexed chains in the ring language; the upstream version should generalize the index order and language — the formula induction carries over verbatim, with the relation case no longer vacuous). Companion: `elementaryInclusion` and `directed_iSup_isElementary`.
4. PR 3 (after dedup check): `Real.isRealClosed`, the root-closure criterion `isRealClosed_of_forall_root`, and `relAlgebraic_isRealClosed`.
5. Long-term: formalize quantifier elimination / model completeness for RCF in Mathlib’s `ModelTheory` framework, modeled on the ACF files but via Robinson’s test, discharging `RCFSubfieldRealElementary`.

## 6. RELATED WORK

### 6.1. Formalizations of the independence of CH.

The set-theoretic background is classical: Gödel proved relative consistency of CH with ZFC, and Cohen proved the independence direction using forcing [1, 2]. Crucially for the present paper, this metatheory has itself been formalized. The **Flypitch** project of Han and van Doorn gave a formal proof of the independence of the Continuum Hypothesis in Lean, with both consistency directions obtained by forcing with Boolean-valued models — Cohen forcing for  $\neg \text{CH}$  and collapse forcing for CH [8, 11]. Independently, Gunther, Pagano, Sánchez Terraf, and Steinberg formalized the independence of CH in Isabelle/ZF [12].

The contrast sharpens the positioning in one sentence: Flypitch and the Isabelle/ZF line formalize the **metatheory** (unprovability, via forcing models), while the present development works **object-theoretically inside** the  $\neg \text{CH}$  regime,

building structure on intermediate cardinalities rather than building models of set theory. The present work contributes nothing to the independence problem itself; it presupposes its resolution and inhabits one of the two resulting worlds.

## 6.2. Real closed fields and certified quantifier elimination.

The algebraic and model-theoretic backbone is the theory of real closed fields. Tarski’s quantifier elimination [3] makes them the natural setting for definability questions involving addition, multiplication, and order [4, 5]; the standard real-algebraic reference is Bochnak–Coste–Roy [13].

On the formalization side, the gap `RCFSubfieldRealElementary` (compatibility spelling `RCFModelComplete`) has substantial prior art in other proof assistants. Cohen and Mahboubi formalized, in `Coq/SSReflect`, a certified quantifier-elimination procedure for the first-order theory of real closed fields, yielding decidability with a correctness proof [9] (the `math-comp/real-closed` library). McLaughlin and Harrison implemented a proof-producing decision procedure for RCF in `HOL Light` [10]. Neither development transfers directly to Mathlib’s deep-embedded `FirstOrder` framework, but Cohen–Mahboubi is the closest blueprint for the remaining gap; Section 5 explains why the cheaper categoricity route used for Mathlib’s ACF completeness is unavailable for RCF.

## 6.3. Formalized mathematics in Lean and Mathlib.

The project belongs to the growing ecosystem of Lean 4 and Mathlib formalizations [14, 15]. Two parts of Mathlib carry the present development: the model theory library (elementary substructures, Skolem functions and downward Löwenheim–Skolem, `Language.DirectLimit`, and the deep-embedded algebra of `Language.ring` up to the completeness of ACF) and the cardinal/ordinal library (König’s theorem, cofinality, fundamental sequences). ICAH is not only a Lean artifact but also a diagnostic of Mathlib boundaries: it identifies concrete places where model theory, real closed fields, and directed colimits need stronger APIs — most pointedly, the absence of any elementarity result for `Language.DirectLimit` and of model completeness for RCF.

## 6.4. Elementary chains and direct limits.

Elementary-chain arguments are standard in model theory; the canonical reference is Chang–Keisler [6], and the König/cofinality background is in Kunen [16] and Jech [17]. The formal contribution here is more specific: it tests how far Mathlib’s `Language.DirectLimit` API can be pushed toward a fully formal elementary-chain theorem. The relativized elementarity theorem (`tarskiVaughtDirectLimit`, via `DirectLimit.lift` and the Tarski–Vaught test), the **ambient-free** chain theorem (`ofLevelElem`: the direct limit of an elementary chain is an elementary extension of each level, by formula induction), and the cardinality identity (`directLimit_card_eq_iSup`) are all proved locally; they are natural candidates for upstreaming, since Mathlib’s `DirectLimit` file currently has no elementarity results.

## 7. NEIGHBORING FIELDS AND ANALOGUES

The architecture instantiates a pattern — carrier, closure property, well-behaved first-order theory, elementary embeddings, direct limits, cardinal bookkeeping — that has natural analogues beyond real closed fields. The closest is the p-adic side:

a p-adically closed analogue would replace subfields of  $\mathbb{R}$  by valued subfields of a p-adic ambient field, real closure by p-adic closure or henselianity, and the ring language by a valued-field language, with Macintyre’s definability theory playing the role of Tarski’s [18, 19]. Henselian valued fields offer a broader comparison class with the same shape, and algebraically closed fields a simpler control case (whose Mathlib model theory is, notably, already complete enough to serve as the template discussed in Section 5). These analogies are deliberately deferred to future work: the core contribution stays focused on the formal Lean development around the real continuum.

## 8. CONCLUSION, REPRODUCIBILITY, AND ENGAGEMENT PLAN

The current formalization is best presented as a rigorous formalization study rather than as a finished foundational theory. Its strongest components are already useful beyond the project name: the DLS-based existence of elementary substrata of  $\mathbb{R}$  at every intermediate cardinality (`exists_elementary_substratum`), the nesting and directed-union lemmas (`elementaryInclusion`, `directed_iSup_isElementary`), a strictly increasing  $\aleph$ -chain (`strictElemChain`), a cofinal elementary family of length  $\text{cof}(c)$  (`exists_cofinal_elem_family_cof_length`), the real-closedness of  $\mathbb{R}$  (`Real.isRealClosed`), the existence of named real-closed subfields of every infinite cardinality up to the continuum (`exists_rc_subfield`), the Tarski–Vaught elementarity theorems for arbitrary countable direct limits — both relativized (`tarskiVaughtDirectLimit`) and ambient-free (`ofLevelElem`) — and the direct-limit cardinality identity and closure theorem (`directLimit_card_eq_iSup`, `directLimit_card_lt_continuum`).

Pillar A formalizes the informal picture under  $\neg \text{CH}$  alone: elementary strata at every intermediate size, a strictly increasing countable chain, and a cofinal elementary family of optimal length. Pillar B remains the conditional, algebraically explicit companion: named real-closed subfields with native `IsRealClosed` instances, whose inclusions into  $\mathbb{R}$  (and into each other) are elementary only given `RCFModelComplete`.

The most important editorial decision is to keep the paper honest about the distinction between proved Lean results and named hypotheses. This is not a weakness: the hypothesis inventory — `NotCH` for the regime, `RCFSubfieldRealElementary` / `RCFModelComplete` for Pillar B only, both visible in type signatures, with the kernel-level audit machine-checked by `#guard_msgs` — turns an ambitious mathematical program into precise, independently attackable formalization problems. Two defects of earlier drafts — a vacuous cardinality theorem and two false subfield axioms, with  $\mathbb{Q}$  as counterexample — were detectable from the development’s own documentation and are now repaired here.

### 8.1. Reproducibility.

The development is pinned and machine-checked:

- **Toolchain:** `leanprover/lean4:v4.31.0-rc2`.
- **Mathlib revision:** `a810615ff479602ad66b5403d179bfa805314a50` (June 2026), locked in `lake-manifest.json`.
- **Release:** this manuscript corresponds to the post-referee Lean revision on the `elementary-strata` branch; a `v1.1.0` tag should be cut after the Zulip dedup check, not the earlier `v1.0.0` artifact.

- **Paper compiler:** Typst 0.14.2, pinned and checked by the paper Makefile and CI.
- **Continuous integration:** lake build (including the guarded audits), zero-sorry and zero-project-axiom checks, followed by a pinned Typst build of the manuscript.
- **Statistics:** 11 Lean modules under ICAH/, circa 2,200 lines of Lean, 0 project axioms, 0 sorries; both main theorems audit to [propext, Classical.choice, Quot.sound].

## 8.2. Declaration-to-paper mapping.

| Lean declaration                             | Paper role                                                                     | Hypotheses |
|----------------------------------------------|--------------------------------------------------------------------------------|------------|
| NotCH                                        | the $\neg$ CH regime (Def., Section 5)                                         | —          |
| RCFSubfieldRealElementary / RCFModelComplete | $\mathbb{R}$ -specialized RCF subfield-inclusion elementarity (Gap, Section 5) | —          |
| exists_intermediate_cardinal                 | intermediate cardinal witness                                                  | NotCH      |
| Stratum, SizeAwareField                      | core objects (Section 3)                                                       | —          |
| syntheticStratum                             | concrete $\aleph_1$ -stratum                                                   | NotCH      |
| exists_elementary_substrata                  | Pillar A attending via DLS                                                     | ZFC-pure   |
| elemSubstratumSubfield                       | substrata are subfields                                                        | ZFC-pure   |
| elemSubstratum_models_thReal                 | substrata model $\text{Th}(\mathbb{R})$                                        | ZFC-pure   |
| elementaryInclusion                          | nested elementary pair                                                         | ZFC-pure   |
| directed_iSup_isElementary                   | directed unions stay elementary                                                | ZFC-pure   |
| strictElemChain                              | strict $\aleph$ -chain witness                                                 | NotCH      |
| exists_rc_subfield                           | Pillar B strata, every $\aleph_0 \leq \kappa \leq c$                           | ZFC-pure   |
| relAlgebraic_isRealClosed                    | relative algebraic closures are RC                                             | ZFC-pure   |
| Real.isRealClosed                            | $\mathbb{R}$ is real closed                                                    | ZFC-pure   |
| fieldOnStratum                               | packaging lemma (Section 4)                                                    | ZFC-pure   |
| tarskiVaughtDirectLimit                      | chain theorem rel. $\mathbb{R}$                                                | ZFC-pure   |
| ofLevelElem / realize_ofLevel_iff            | ambient-free chain theorem                                                     | ZFC-pure   |
| directLimit_card_eq_iSup                     | limit cardinality identity                                                     | ZFC-pure   |

| Lean declaration                                          | Paper role                                          | Hypotheses              |
|-----------------------------------------------------------|-----------------------------------------------------|-------------------------|
| <code>directLimit_card_lt_cont</code>                     | closure under countable chains                      | ZFC-pure                |
| <code>cofinal_family_length_lower_bound</code>            | covering subsets, $\text{cof}(c)$ lower bound       | ZFC-pure                |
| <code>exists_cofinal_elem_family</code>                   | $\omega$ -indexed elementary covering family        | NotCH                   |
| <code>exists_cofinal_elem_family_of_length</code>         | $\text{cof}(c)$ -length elementary covering family  | NotCH                   |
| <code>exists_cofinal_rc_family</code>                     | $\omega$ -indexed real-closed covering family       | NotCH                   |
| <code>exists_cofinal_rc_family_of_length</code>           | $\text{cof}(c)$ -length real-closed covering family | NotCH                   |
| <code>exists_cofinal_rc_family_Elementary</code>          | Elementary with elementary inclusions               | NotCH, RCFModelComplete |
| <code>cofinal_family_limit_size</code>                    | M6 limit-size clause                                | NotCH                   |
| <code>ICAHElementary</code> / <code>icahElementary</code> | <b>Theorem 1</b> (Pillar A)                         | NotCH                   |
| <code>ICAHStatement</code> / <code>icahTheorem</code>     | <b>Theorem 2</b> (Pillar B)                         | NotCH, RCFModelComplete |

### 8.3. Engagement plan.

1. Post on the Lean Zulip model-theory stream with the repository and a minimized in-framework statement of `RCFSubfieldRealElementary`; this may surface in-progress RCF or real-closure work and prevents duplicated upstreaming. (A draft is maintained in `docs/UPSTREAMING.md`.) Do not claim this is the unique Mathlib gap until that check returns.
2. PR sequence: (1) DLS-based elementary substrata of  $\mathbb{R}$  plus `directLimit_card_eq_iSup` — small, ZFC-pure, attractive; (2) the ambient-free Tarski–Vaught chain theorem for `Language.DirectLimit`, together with `elementaryInclusion` and `directed_iSup_isElementary`; (3) `Real.isRealClosed`, `isRealClosed_of_forall_root`, `relAlgebraic_isRealClosed` after the dedup check; (4) long-term, RCF model completeness modeled on the ACF files via Robinson’s test.
3. Venue: this is a formalization paper — ITP or CPP, with an arXiv preprint cross-listed cs.LO primary and math.LO secondary. It should not be sent to a set-theory venue as-is, where the mathematics reads as exercises. JAR is appropriate only if the RCF-gap analysis is substantially expanded.
4. Keep p-adic and valued-field analogies as future work, not as the main thesis.

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